

# Computer Networking Basics

A Test Document for PDF Question-Answering Systems

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| <b>Purpose</b>              | Sample theory + Q&A content to test a PDF-reading QA project |
| <b>Sections</b>             | 5 theory topics, each followed by review questions           |
| <b>Answer key</b>           | Included at the end of the document                          |
| <b>Word count (approx.)</b> | ~1,400 words                                                 |

# 1. What Is a Computer Network?

A computer network is a collection of two or more devices, called nodes, that are connected together to share resources and exchange data. These nodes can be computers, servers, printers, routers, or mobile devices. The connections between nodes may be established using physical media such as copper cables and fiber optics, or through wireless media such as radio waves and infrared signals. The main purpose of a network is to allow devices to communicate with each other, share files, access shared hardware like printers, and connect to broader networks such as the internet.

Networks are typically classified by their geographical scope. A Local Area Network (LAN) connects devices within a small area such as a home, office, or school building. A Metropolitan Area Network (MAN) spans a larger area such as a city. A Wide Area Network (WAN) connects devices across large geographical distances, sometimes spanning countries or continents; the internet itself is the largest example of a WAN. Choosing the right type of network depends on factors such as distance, required speed, and cost.

## Review Questions - Section 1

Q1. What is a computer network?

Q2. Name the three types of networks classified by geographical scope.

Q3. Give one example of physical media used to connect network nodes.

# 2. The OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework that standardizes the functions of a communication system into seven distinct layers. Starting from the bottom, these layers are: Physical, Data Link, Network, Transport, Session, Presentation, and Application. Each layer serves the layer above it and is served by the layer below it, which allows different technologies and vendors to interoperate as long as each layer follows the standard rules.

The Physical layer deals with the transmission of raw bits over a physical medium. The Data Link layer organizes bits into frames and handles error detection between directly connected nodes. The Network layer is responsible for logical addressing and routing packets between different networks, and this is where the Internet Protocol (IP) operates. The Transport layer ensures reliable or unreliable end-to-end delivery of data, with TCP providing reliable delivery and UDP providing faster but unreliable delivery. The Session, Presentation, and Application layers handle higher-level concerns such as maintaining conversations, formatting or encrypting data, and providing services directly to end-user software.

## Review Questions - Section 2

Q4. How many layers does the OSI model have, and what is the bottom-most layer called?

Q5. Which OSI layer is responsible for routing packets between networks?

Q6. What is the key difference between TCP and UDP at the Transport layer?

### **3. IP Addresses and Subnetting**

An IP address is a unique numerical label assigned to each device connected to a network that uses the Internet Protocol for communication. IPv4 addresses consist of 32 bits, typically written as four decimal numbers separated by dots, such as 192.168.1.1, where each number ranges from 0 to 255. Because the pool of IPv4 addresses is limited, IPv6 was introduced, using 128-bit addresses to provide a vastly larger address space.

Subnetting is the practice of dividing a larger network into smaller sub-networks, or subnets, to improve performance, security, and address management. A subnet mask determines which portion of an IP address refers to the network and which portion refers to the host. For example, a subnet mask of 255.255.255.0 means the first three octets identify the network, leaving the last octet to identify individual hosts within that subnet, allowing up to 254 usable host addresses.

#### **Review Questions - Section 3**

Q7. How many bits does an IPv4 address contain?

Q8. Why was IPv6 introduced?

Q9. With a subnet mask of 255.255.255.0, how many usable host addresses are available?

## 4. DNS and How Websites Are Found

The Domain Name System (DNS) is often described as the phonebook of the internet. It translates human-readable domain names, such as example.com, into numerical IP addresses that computers use to identify each other on the network. Without DNS, users would need to memorize IP addresses for every website they wanted to visit.

When a user types a domain name into a browser, the device first checks its local cache for a matching IP address. If not found, it queries a DNS resolver, which in turn may contact root servers, top-level domain (TLD) servers, and finally the authoritative name server for that domain to retrieve the correct IP address. This entire lookup process usually takes only a fraction of a second and happens transparently in the background before the browser requests the actual web page content from the server.

### Review Questions - Section 4

Q10. What does DNS stand for, and what is its main function?

Q11. What does a device check first before sending a DNS query to a resolver?

## 5. Firewalls and Network Security

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules. Firewalls act as a barrier between a trusted internal network and untrusted external networks, such as the internet, blocking traffic that does not meet the configured rules while allowing legitimate traffic to pass through.

There are several types of firewalls. Packet-filtering firewalls examine individual packets and allow or block them based on source and destination addresses, ports, and protocols. Stateful inspection firewalls track the state of active connections and make decisions based on the context of the traffic, not just individual packets. Application-layer firewalls, sometimes called proxy firewalls, inspect the actual content of traffic at the application level, allowing them to detect threats that simpler firewalls would miss.

### Review Questions - Section 5

Q12. What is the primary function of a firewall?

Q13. Name the three types of firewalls described above.

## Answer Key

- A1.** A computer network is a collection of two or more connected devices (nodes) that share resources and exchange data.
- A2.** LAN (Local Area Network), MAN (Metropolitan Area Network), and WAN (Wide Area Network).
- A3.** Copper cables or fiber optic cables (wireless media such as radio waves is also acceptable).
- A4.** Seven layers; the bottom-most layer is the Physical layer.
- A5.** The Network layer.
- A6.** TCP provides reliable delivery of data, while UDP provides faster but unreliable delivery.
- A7.** 32 bits.
- A8.** Because the pool of available IPv4 addresses is limited; IPv6 provides a much larger 128-bit address space.
- A9.** 254 usable host addresses.
- A10.** DNS stands for Domain Name System; its main function is translating domain names into IP addresses.
- A11.** It checks its local cache for a matching IP address.
- A12.** To monitor and control incoming and outgoing network traffic based on security rules, acting as a barrier between trusted and untrusted networks.
- A13.** Packet-filtering firewalls, stateful inspection firewalls, and application-layer (proxy) firewalls.